

Retail Sales Analytics – SQL Project

1. Project Overview

This project analyzes retail sales data using SQL Server. The goal is to understand sales performance, customer behavior, product performance, and order trends.

The project was created using SQL Server and includes database creation, table creation, data insertion, analysis, and KPI calculations.

2. Project Objective

The main objectives of this project are:

- Analyze total revenue and sales performance
 - Identify the highest-revenue product categories
 - Find the top-selling products
 - Analyze customer purchasing behavior
 - Calculate Average Order Value
 - Analyze order cancellations
 - Understand monthly revenue and growth trends
 - Create important business KPIs
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3. Business Questions

The project answers the following business questions:

1. What is the total revenue from completed orders?
2. Which category generates the highest revenue?
3. Which products sell the most units?
4. Which customers generate the highest revenue?
5. What is the Average Order Value?
6. How does monthly revenue change over time?

4. Database Tables

The project contains four main tables:

Customers

Contains customer information such as:

- Customer ID
- First Name
- Last Name
- City

Products

Contains product information such as:

- Product ID
- Product Name
- Category

Orders

Contains order information such as:

- Order ID
- Customer ID
- Order Date
- Order Status

Order_Details

Contains details about products in each order:

- Order ID
- Product ID
- Quantity
- Unit Price

5. Tools Used

- SQL Server
- SQL Server Management Studio (SSMS)

6. SQL Skills Used

- CREATE DATABASE
 - CREATE TABLE
 - INSERT
 - SELECT
 - WHERE
 - JOIN
 - LEFT JOIN
 - GROUP BY
 - ORDER BY
 - TOP
 - COUNT
 - SUM
 - AVG
 - ROUND
 - CAST
 - DISTINCT
 - CASE
 - YEAR
 - MONTH
 - Common Table Expressions (CTEs)
 - LAG()
 - Revenue growth calculation
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7. Key KPIs

KPI	Result
Total Orders	40
Completed Orders	30
Cancelled Orders	4
Total Units Sold	81
Total Revenue	\$5,834.9
Average Order Value	\$194.47
Unique Customers	15
Cancellation Rate	10.00%

8. Analysis Results

8.1 Total Revenue

The total revenue from completed orders is:**\$5,834.19**

Only completed orders were included when calculating revenue.

8.2 Revenue by Category

Category	Revenue
Furniture	\$2,349.87
Electronics	\$1,619.71
Home & Kitchen	\$1,279.79
Accessories	\$399.92
Office Supplies	\$184.90

Furniture generated the highest revenue with **\$2,349.87**.

8.3 Top 5 Products by Units Sold

Product	Category	Units Sold
Wireless Mouse	Electronics	9
Desk Organizer	Office Supplies	7
USB-C Hub	Electronics	7
Electric Kettle	Home & Kitchen	6
Bluetooth Headphones	Electronics	5

The **Wireless Mouse** was the top-selling product with **9 units sold**.

8.4 Highest Revenue Customer

The customer with the highest revenue was:

Sophia Anderson (Customer 6)

Total revenue generated: **\$689.90**

8.5 Average Order Value

The Average Order Value was: **\$194.47**

This represents the average revenue generated from each completed order.

8.6 Cancellation Rate

The cancellation rate was: **10.00%**

There were **4 cancelled orders out of 40 total orders**.

8.7 Monthly Revenue

Monthly revenue was analyzed using the order date.

The highest monthly revenue was:

May 2025 — \$489.96

The largest monthly growth was:

May 2025 — 226.71%

The largest monthly decline was:

June 2025 — -70.41%

The first month in the analysis has no previous month available, so its growth percentage is NULL.

9. Key Findings

Based on the analysis:

- Total completed-order revenue was **\$5,834.19**.
 - **Furniture** was the highest-revenue category.
 - **Wireless Mouse** was the top-selling product by units.
 - **Sophia Anderson** generated the highest customer revenue at **\$689.90**.
 - The Average Order Value was **\$194.47**.
 - The cancellation rate was **10%**.
 - **May 2025** had the highest monthly revenue and the highest monthly growth.
 - Revenue showed significant month-to-month fluctuations.
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10. Business Recommendations

1. Focus on High-Performing Categories

Furniture generated the highest revenue. The business can continue focusing on this category and identify similar products that can increase sales.

2. Promote Best-Selling Products

The Wireless Mouse was the top-selling product. Similar high-demand products can be promoted through offers or product bundles.

3. Increase Average Order Value

The business can use product bundles, cross-selling, and related-product recommendations to increase the average order value.

4. Reduce Order Cancellations

The cancellation rate is 10%. The business can review the reasons for cancellations and improve the ordering and fulfillment process.

5. Monitor Revenue Fluctuations

Monthly revenue had significant increases and decreases. Monitoring these changes can help identify strong and weak sales periods.

11. Project Structure

Retail-Sales-Analytics-SQL

```
|  
|  
| └─ SQL  
|   |  
|   | └─ 01_Create_Database.sql  
|   |  
|   | └─ 02_Create_Tables.sql  
|   |  
|   | └─ 03_Insert_Data.sql  
|   |  
|   | └─ 04_Analysis.sql  
|   |  
|   | └─ 05_KPIs.sql  
|  
|  
| └─ Documentation  
|   |  
|   | └─ Project_Documentation.md  
|  
|  
└─ Images
```

12. Conclusion

This project demonstrates how SQL can be used to analyze retail sales data and answer business questions.

The analysis covered revenue, products, categories, customers, orders, cancellations, Average Order Value, and monthly revenue trends.

The project also demonstrates practical SQL skills such as joins, aggregations, filtering, grouping, CTEs, and window functions.